When Work No Longer Suffices to Establish Human Worth

Emergent Intelligence, Social Recognition, and a Crisis of Meaning in Intergenerational Futures

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Abstract If technological progress can reduce the necessity of certain forms of labor, while individuals still depend on work for income, social recognition, and direction, why might greater productive capacity also produce a sense of being superfluous? This paper connects changes in generative AI capabilities, questions of value in the postmodern condition, and expectations concerning future generations. It combines problem-oriented literature analysis, conceptual distinctions, thought experiments, and normative argument, using a completed reanalysis of public data as bounded empirical material. It distinguishes income loss, impaired practical agency, diminished social standing, and disrupted narratives of the future, arguing that technological effects on meaning are mediated by institutions and relationships. In relation to low fertility, the proposed pathway from diminished work-related worth through difficulties imagining a viable future to changing fertility intentions deserves investigation. Existing task experiments do not measure fertility, however, and demographic evidence does not justify interpreting all lower fertility as nihilism. Economic constraints, care arrangements, opportunity costs, preferences, and reverse causation are competing explanations. The paper further distinguishes basic dignity, which should not depend on performance, from meaning in life, which involves concrete engagement. Society cannot guarantee equal achievement, but it has responsibilities concerning the conversion of technological displacement into a person's overall devaluation. Human meaning may arise through participation in a shared world, understanding, relationships, and worthwhile action. This account addresses objections concerning uncompensated losses, merely symbolic participation, and the possible value of future artificial subjects. It offers revisable propositions and open questions rather than a unified explanation of declining birth rates or a final purpose for humanity.

1. The starting problem: why might someone relieved of a burden feel superfluous?

Imagine a professional with ten years of experience. Through sustained training, this person has developed judgment, completed complex tasks through their own understanding, and learned why colleagues trust them and clients need them. AI subsequently makes similar tasks easier. The professional still has an income and submits more work each day, yet finds it increasingly difficult to explain which decisions are their own, which capabilities remain worth developing, and why anyone will seek them out several years from now. This constructed scenario does not describe every worker. It identifies a problem that employment counts alone cannot answer: a job may survive while the relationship between a person and their work changes.

Now suppose that technological progress does make certain jobs unnecessary. Affected people receive two messages at once: society celebrates greater efficiency while continuing to expect everyone to work, improve themselves, and demonstrate their value. What production saves is labor time; those required to justify

themselves anew are the workers whose labor has been saved. Their confusion should not immediately be attributed to a failure to understand progress. They may be accurately experiencing a conflict between two sets of rules: one reduces demand for their labor, while the other continues to make access to work an important condition of respect and a viable future.

This conditional contradiction is the subject of the paper. Productivity growth does not necessarily reduce aggregate employment. AI may create tasks, expand demand for services, and increase individual capabilities. The argument does not assume that everyone will eventually become unemployed. It asks what makes task displacement, changes in professional standing, or persistent expectations of replacement feel liberating in some circumstances and excluding in others. The problem can arise even when total employment remains stable and average incomes increase.

It can also extend through time. Someone who struggles to believe that present effort leads to a future they can reasonably navigate may reconsider training, migration, partnership, and raising the next generation. This leads to a second question: might a weakening connection between work and worth affect reproductive decisions through people's understanding of the future? This is not a direct leap from AI to birth rates. It concerns how people translate their circumstances into expectations about the lives of future children. Precisely because such a leap is tempting, the paper examines the evidence it would require and the explanations that compete with it.

The discussion cannot end by observing that people can find meaning elsewhere. The existence of other worthwhile activities does not make ten years of occupational commitment readily exchangeable. Continued possession of dignity does not mean that someone has recovered a direction in life. A credible account of human meaning must acknowledge the specificity of loss and explain the social conditions under which meaning becomes possible. Otherwise, telling someone that they still have value is correct but unresponsive to their predicament.

The intended contribution is to connect debates on automation and achievement, social recognition, and demographic narratives of the future while separating their links and their distinct evidential requirements. The paper proposes a revisable interpretation: a crisis of meaning may arise when the capacities production requires become disconnected from the ways society recognizes people; hesitation about intergenerational futures may be one expression of that disconnection. This is not a diagnosis of humanity's collective psychology, and lower fertility is not treated as a psychological symptom that requires correction.

2. An argumentative route: distinguishing the meanings of meaning

2.1 Four questions cannot be answered in the same way

Asking about a person's value can involve at least four questions. A market asks whether labor will be purchased and what income it commands. Ethics asks whether someone deserves respect and protection. Social relationships concern whether their views are taken seriously and their contributions noticed. Personal life concerns whether experiences form an intelligible story and whether there remain worthwhile goals. Income, dignity, standing, and meaning interact, but cannot be measured by the same ranking. Lower income need not eliminate all meaning; verbal recognition of dignity need not provide an effective voice.

Psychological research offers further distinctions. Martela and Steger separate the intelligibility of life, its direction, and the sense that it is worth living. This cautions against reducing meaning to pleasure or job satisfaction. The present discussion accordingly distinguishes narrative coherence, purposive direction, and experienced significance. This is an analytical borrowing, not a claim that a new measurement instrument has been administered or validated. [1]

Here, a sense of meaninglessness concerns difficulty finding reasons sufficient to sustain engagement. Value nihilism is a philosophical position about whether values hold. Employment anxiety, occupational dissatisfaction, clinical depression, and value nihilism require different explanations and cannot serve as diagnoses of one another. Similarly, the postmodern is not a label applied to everyone living today. It is a

limited perspective for examining doubts about unified foundations of value and the plurality of reasons for living.

2.2 What was actually done

The study uses problem-oriented literature searches and argument. It is neither an exhaustive systematic review nor a meta-analysis that pools estimates. Materials include research on AI capabilities and work, theories of meaning and recognition, demographic research on future narratives, and an earlier reanalysis of data made public by their original authors. Searches prioritize original papers, data, and institutional reports, while including null results, favorable effects, and alternative explanations. The accompanying evidence register states how deeply each source was checked and which claims it supports. Sources checked only at abstract level do not carry detailed methodological conclusions.

The argument first examines how work becomes evidence of worth, then asks what AI changes and distinguishes pathways from technological change to experienced meaning. Fertility enters after these pathways have been specified. The final sections address normative responses and propositions that could be challenged by evidence. Empirical material limits explanations, thought experiments distinguish concepts, and normative argument addresses how people should be treated. These forms of inquiry constrain rather than substitute for one another. No questionnaire directly establishes humanity's ultimate meaning, and no philosophical principle independently estimates a technology's employment or demographic effects.

3. How work becomes a way of establishing one's worth

3.1 Social standing beyond wages

The need to work to sustain life does not imply a requirement to establish one's entitlement to exist through paid employment. The first claim concerns production and distribution; the second converts a market position into a moral assessment. They are easily conflated because occupations often organize income, time, social contact, skill development, and the ways others introduce a person. When an occupation commonly answers the question of who someone is, a job provides an intelligible social position as well as remuneration.

Society's demands do not originate from a single coordinated will. Employers pursue performance and cost control; educational institutions promise better opportunities through learning; families expect stable occupations; some public arrangements organize protection around employment status; and individuals internalize these expectations. Actors with different motives may reinforce a common life pattern without anyone being able to guarantee its continued viability. This dispersed structure does not make its consequences unreal. It helps explain why displaced workers may find no single party responsible for the wider promise on which they relied.

Work as evidence of worth has genuine attractions. Receiving payment and trust for doing something others need can establish that one participates in collective life rather than merely being tolerated. The problem arises when this becomes an exclusive standard. Care, learning, recovery from illness, civic participation, and nonmarket creativity can then be treated as value not yet realized, receiving full recognition only when recorded as wages or performance. Markets do not register all social needs, and purchasing power does not track their urgency. Reduced market demand therefore cannot establish a reduction in a person's overall worth.

3.2 Why liberation from labor can become a loss of identity

In the third volume of Capital, Marx relates necessary labor to freedom and identifies a shorter working day as a condition of free development. The argument does not eliminate the need for production and care. It asks whether people can collectively control productive conditions so that sustaining life does not exhaust life. This raises an institutional question for the present paper: who controls the time technology saves, and who receives the resulting gains? If a few acquire greater choice while others lose livelihoods or standing, increased productive capacity will not automatically become freedom for all. [2]

The difference between losing work and escaping compulsory labor depends on whether people retain resources, choices, and relationships. Someone who reduces working hours with material security and control over their time may experience something very different from someone made involuntarily idle by dismissal. Both work less, but describing both situations simply as more leisure removes the conditions that shape their meaning.

Nor should earlier forms of work be romanticized. Repetitive, dangerous, or humiliating work does not deserve preservation merely because it demonstrates diligence. Requiring some people to continue unnecessary suffering in order to retain a sense of achievement turns hardship into a test of deservingness. The difficult task is to reduce labor that should not have to be endured without treating those who performed it as remnants to be cleared away.

3.3 Task substitution and human replaceability

Most positions in a division of labor could already be filled by someone else. If only irreplaceable people had meaning, most people would have lacked it before AI. Ordinary work often matters because it actually improves someone's circumstances and develops skills or cooperation, not because nobody else in the world could do it. The problem requiring explanation is therefore whether routes through which people could make a difference are being closed, and whether accessible alternatives are opening.

This distinction also limits pessimistic conclusions. Some displaced people enter more valuable practices or discover that their occupation was never their principal source of meaning. Others face obstacles involving age, care responsibilities, local opportunities, and resources. A shared technological event does not produce a uniform life outcome. Research should examine who can move, what the transition costs, and which losses new opportunities cannot repair, rather than requiring everyone to adopt the same attitude toward technology.

3.4 Why a wealthier society can coexist with households finding planning harder

Higher productivity means more can be produced per unit of input. It does not determine who receives the gains through wages, profits, lower prices, or public services. Even when some goods become cheaper, a household with insecure earnings may hesitate to make long-term commitments. Greater aggregate output need not relieve housing, care, or education constraints proportionately. Technological enrichment and growing difficulty planning within some households can therefore coexist. Distribution and the organization of services connect the two.

Labor demand also changes in more than one way. Producing a fixed quantity may require fewer people, while lower prices, additional demand, and new tasks can expand employment. Which force dominates is empirical. For particular workers, average gains may arise outside occupations they can enter. The contradiction examined here is not that greater productivity necessarily makes life worse. It is that the creation of technological gains and people's eligibility to receive them may be governed by different rules.

This shifts attention from whether individuals are optimistic enough to how transitions are organized. Caution may be reasonable where workers and families carry risks without a corresponding share of gains. Describing it as insufficient ambition obscures its institutional context; describing every cautious choice as institutional harm overlooks differences in people's wishes. Research must preserve both possibilities.

3.5 Why a society without necessary employment could remain organized around work

Reducing hours of labor does not necessarily reduce labor's authority to define a person. Someone with adequate material support may still judge themselves by former occupational rankings and experience an unoccupied day as a loss of standing. At the opening of "Labor, Work, Action," Arendt distinguishes activities that sustain life, provide a human dwelling, and organize collective life. This distinction cautions against exhausting active life in the institutional form of paid employment. The source used here is the publicly available opening preview, not a reconstruction of her entire political philosophy. [3]

The difficulty is that an evaluative order may outlive the productive need that helped sustain it. Imagine a city able to give residents an additional day free of employment while still treating constant availability, immediate responses, and continuous promotion as marks of responsible adulthood. The new time could be absorbed by training, credential competition, and self-presentation. Productive necessity would diminish while evaluative necessity persisted: people might become busier demonstrating that they remain necessary. This is a thought experiment about potentially offsetting arrangements of time, not an empirical report about a city.

An easily overlooked object of distribution is freedom from the burden of justification. People with greater standing may describe free time as exploration, recovery, or reflection; others may be expected to explain the same time as preparation for employment or idleness awaiting correction. Where that difference operates, time saved by technology is not equally available to everyone. Freedom involves more than gaining hours. It also concerns whether a person must continually explain to someone more powerful how those hours will eventually become output.

Complete exemption from accountability would also be unreasonable. Shared life requires care, maintenance, and public responsibilities, and people may legitimately ask whether burdens are fairly distributed. The relevant question should move from how busy someone appears to how necessary responsibilities are shared, who decides, and which activities need not justify themselves through productive returns. This neither guarantees a wish to leave employment nor abolishes the value of effort. It rejects treating a historically formed organization of life as an eternal human essence.

3.6 What separates being valued from having bargaining power

Even universal agreement that human worth cannot be discounted could coexist with deeply unequal relationships. Someone may be treated politely and repeatedly praised yet be unable to reject unfavorable arrangements because another party controls their resources. Recognition of worth is a normative requirement; being able to say no requires practical conditions. The distance between them is precisely what reassuring language about meaning can conceal.

The market value of labor has given some people a bargaining position: an organization needs their cooperation and must therefore respond to their demands to some degree. If technology weakens that need without corresponding changes to protection and voice, verbal affirmations of dignity may not preserve their standing. Concern about this cannot be reduced entirely to vanity. What someone wishes to retain may be the conditions of receiving serious consideration without depending on another party’s goodwill, rather than a feeling of superiority.

The inference that work has become unnecessary and people are therefore free is consequently premature. Freedom from having to sell labor to survive can be an advance. An inability to influence common rules because one has nothing to sell can become another dependency. Whether technology reduces compulsion requires attention to the sources of income, the possibility of arbitrary withdrawal of support, and opportunities to organize and object. Meaning is not another name for bargaining power, but weak bargaining conditions can allow others to prearrange the available choices about meaning.

4. Why emergent intelligence makes an older problem newly pressing

4.1 Capabilities, thresholds, and an unsettled boundary

Research on emergent abilities describes improvements on some tasks that are difficult to extrapolate from smaller models. Research on apparent emergence argues that some abrupt changes arise from discontinuous evaluation metrics. This debate requires a distinction between capabilities and measured performance; it does not settle consciousness or strong emergence. For workers, another threshold matters: whether a system can perform part of a real task at an acceptable cost and error rate. A smooth capability curve can still cross that social threshold. [4], [5]

The paper does not infer subjective experience from fluent output. Capability, consciousness, and moral status are related but distinct questions. Work deriving structural and functional indicators from consciousness theories illustrates why behavioral impressions alone are insufficient. Nor are assessments of earlier systems used to determine the status of every later system. Even if future artificial subjects have experiences, deciding how they should be treated would not establish that humans cease to deserve concern. [6]

AI places particular pressure on work-related meaning by entering activities through which people commonly understand their professional capabilities: writing, synthesis, programming, design, judgment, and proposing solutions. Systems vary in reliability across these activities. Nevertheless, when boundaries keep shifting, the assurance that machines cannot do a particular thing becomes an unstable occupational promise. The issue concerns both present substitution and whether a person can trust that another long period of learning will lead to a durable place.

4.2 The end of work cannot be assumed

The ILO's 2025 occupational exposure assessment estimates that approximately one in four workers worldwide is in an occupation with some generative AI exposure, while 3.3% of employment falls in the highest exposure category. It also emphasizes that most occupations contain tasks requiring human input, making transformation more likely than complete replacement. Exposure measures potential task effects, not an individual's probability of unemployment. The one-in-four estimate must not be read as a proportion of jobs about to disappear. [7]

Demand also depends on whether lower prices expand services, organizations develop new tasks, additional review becomes necessary, and human-system roles are rearranged. Technology can replace one step while enabling a person to complete an entire project for the first time. The focus here is not a predetermined endpoint without human labor, but an uneven transition: a technology can benefit aggregate production without preserving opportunities for contribution, learning, and bargaining for everyone affected.

4.3 Explaining intelligence does not cancel meaning

Intelligence arising in artificial systems may unsettle a deeper self-understanding: if thought, emotion, and choice can all be explained through mechanisms, are people merely complex computations? This question compresses several different issues: whether we understand how a capacity arises, whether we have fully explained a subject's experience, and whether that explanation weakens the reasons for taking the subject's life seriously. Advances in generating capabilities do not directly settle the latter two questions. Even if explanations become more complete, the step from understanding a mechanism to cancelling meaning still requires an argument.

First, explaining origins and judging value perform different argumentative tasks. In A Treatise of Human Nature, Hume asks for an explanation of the transition from statements about what is to statements about what ought to be. Borrowing this question does not require treating facts and values as unrelated. Knowing what harm someone is undergoing plainly affects how we should act. Yet the mechanism producing that harm does not automatically supply a reason to permit it. Explaining how suffering occurs and judging whether it should be alleviated must be connected without being treated as interchangeable. [8]

The claim that emotions have a biological basis and therefore lack value consequently depends on a premise that needs defense: why should only things without natural causes deserve concern? Under that standard, the more we understand, the fewer objects of concern remain, leaving human dignity lodged wherever knowledge has not yet reached. Scientific explanation does not itself supply this standard, which would make meaning depend on ignorance. The reverse inference also fails: a naturally occurring impulse does not necessarily deserve obedience. Jealousy and care may both have explicable causes; whether to support them still requires considering how they treat others, respond to facts, and bear consequences.

A description of a process must also be distinguished from what that process is like for its subject. Nagel's discussion of subjective experience asks accounts of consciousness to confront the experiential point of view. This paper borrows that question without deciding whether physicalism is true or asserting that machines

can never have experience. If a claim to have explained people actually concerns only behavioral prediction or task performance, it should not silently count all experience and meaning as already explained. The limitation concerns whether the object of explanation has been adequately included, rather than reserving an uninvestigable mystery for humanity. [9]

Even if we temporarily adopt a stronger stipulation and suppose that a future theory fully explains how human experience arises, questions of value do not thereby disappear. Knowing every condition producing an episode of loss does not mean that the loss never occurred. Explaining why someone cares about a promise does not mean that breaking it cannot harm them. Explained experience remains an object requiring evaluation. Better knowledge may allow more accurate identification of harm, correction of misunderstanding, or improvement of care, rather than automatically reducing concern. What is preserved here is the need for judgment after explanation, not a claim that experience must forever remain inexplicable.

A similar confusion concerns reproducibility and importance. If understanding can arise through different routes, or a capacity becomes available to more subjects, its scarcity may decline without directly cancelling the value of particular instances for their participants. Someone who learns to express their needs has not gained nothing merely because others already possess that ability. Exclusive market advantage and whether a condition of life is worth having are different objects of evaluation. If AI makes some capabilities more widely available, it may first change the distribution of advantages. A further argument is needed to conclude that human lives consequently lose meaning as a whole.

Meaning nevertheless cannot escape criticism merely because an experience really occurred. Someone may feel accomplishment through a mistaken belief that they helped others, then discover that their work failed to provide the intended help or even caused harm. Explanation can then undermine a judgment of meaning by revealing that its factual grounds were false. If loyalty is sustained principally through deception, uncovering its formation may also change how the relationship should be assessed. Explaining how an experience arises must be distinguished from showing that it depends on error, manipulation, or harm. The latter can warrant revision and must not be dismissed merely by separating value from mechanism.

In responding to AI-related changes in self-understanding, we should therefore neither shelter meaning from explanation nor read every explanation as exposure of an illusion. The specific questions are which factual belief new knowledge overturns, which inequality it reveals, or whether it merely identifies the conditions under which a valuable activity arises. If someone values care because they believe it actually helps its recipient, discovering contrary facts matters. If the discovery instead concerns the psychological mechanisms through which they care about others, it does not by itself establish that the concern is false. Whether meaning can withstand revision deserves more attention than whether it remains mysterious.

Choice requires similar caution. If decisions have causal conditions, does that mean every choice is merely a predetermined process? This paper does not claim to settle the free-will debate. It adopts a narrower practical distinction: being able to understand reasons, access relevant information, respond to objections, and change one's actions differs from being deceived, coerced, or deprived of conditions for exit. Actions with causal explanations can still differ in these respects. Saying that a decision has causes does not erase those differences. Conversely, unpredictability alone cannot establish autonomy, since disorder and insufficient information can also defeat prediction.

Consider a system that accurately predicts a worker's agreement to overtime. This is a constructed scenario, not a claim about the capacities of existing systems. In one case, the worker understands the arrangement, can refuse it, and endorses their reasons for undertaking the work. In another, the organization exploits pressures the worker has not recognized, conceals alternatives, and cultivates an expectation of exclusion for noncompliance. The predicted outcome is the same, but the conditions of participation differ. Better predictive accuracy cannot substitute for judging those conditions. The relevant question is whether knowledge about a person expands their understanding and choices or circumvents their judgment.

This distinction returns an apparently abstract philosophical problem to social relationships. If an organization takes its capacity to predict people's responses as a reason to stop listening to their reasons, people

may become objects awaiting optimization rather than participants in decisions. Harm need not arise from the accuracy of an algorithmic explanation. It may arise when explanatory authority is converted into power over others: a person is described in increasing detail while losing opportunities to say what they accept. Technology can assist judgments about achieving a goal, but who may set that goal and whether those affected can object remain institutional questions requiring justification. Knowing how to influence someone does not establish authority to arrange their life for them.

If future artificial subjects can also experience, understand, and form relationships, why should humans matter? This objection deserves serious consideration, but value must not be assumed in advance to be a position reserved for the strongest. Among humans, slower understanding or weaker expression should not make someone's suffering or wishes negligible. If that reason holds, the appearance of more capable artificial subjects does not make it suddenly fail. Extending moral concern does not logically revoke the standing of subjects already within its scope. Limited resources and conflicting interests still require particular judgments. This is a normative commitment requiring defense, rather than a scientific conclusion directly deduced from the mechanisms of intelligence.

Losing the assurance that humans will always be the most intelligent may nevertheless create real difficulties of self-understanding. Someone may have invested years in training not only for income but because they believed their judgment had a place in the world. Being told that value has not been logically cancelled cannot immediately repair that loss. Opportunities to learn, rules of participation, and how others respond to their experience may also need to change. The paper resists conflating the loss of superior capability with the loss of every reason for living. It also recognizes that relationships and institutions must support this distinction if it is to become a livable reality.

The question thus moves from proving that people possess some property machines can never acquire to asking how increasingly explicable people can still be treated as subjects who experience, respond, and participate in shared life. Human meaning cannot be protected solely through cognitive exceptionalism or sustained by a single verbal affirmation. The knowledge explanations provide must ultimately face a practical question: does it improve people's conditions for understanding their circumstances and participating in decisions, or make it easier for external systems to perform those activities in their place? This leads into the next section: when capabilities can be accessed on demand, does the process of developing judgment still belong to the participant?

4.4 Capabilities can be accessed on demand; can judgment be outsourced with them?

AI can make capabilities available without the individual always undergoing prolonged training. This may widen opportunity while raising a further question: has someone acquired an ability, or access to an ability supplied elsewhere? The distinction is not absolute. Schools, libraries, software, and colleagues have always supported what individuals can do; entirely unsupported competence does not exist. Different dependencies nevertheless provide different degrees of understanding and control.

Consider two constructed learning processes. In one, a system supplies answers while helping the learner understand reasons, detect errors, and gradually assume more judgment. In the other, it continuously delivers answers while understanding is treated as an unnecessary cost. Immediate products could resemble one another while the people formed through these processes could differ. This does not establish that either process inevitably causes deterioration. It shows why evaluation confined to the current product can overlook whether capabilities accumulate.

If dependable judgment increasingly comes from a small set of systems, people may face another vulnerability: retaining the conditions for recognizing when a system should not be trusted. Those conditions include knowledge, time, alternative sources, and organizational permission to object. Asking individuals to remain critical while removing the training and authority that make criticism possible assigns a responsibility whose practical basis has been withdrawn.

Conversely, preserving human training does not justify imposing preventable harm on those served. A learner's development matters, but that does not confer a right to reject more reliable assistance in every situation. What deserves protection is a sustainable arrangement for learning and judgment, rather than a ritual requiring every operation to be performed manually. Education, review, and practical opportunities can be reorganized. Yet if every efficiency gain removes opportunities for future learners, growing present capabilities could coexist with diminished capacities for participation in the next generation.

The social question of emergent intelligence thus has an intergenerational dimension. It concerns not only whether people today are displaced but how people tomorrow become capable of responsible judgment. If humans retain only the role of consuming answers, increasingly good answers could accompany reduced participation in deciding how knowledge is warranted and which goals to pursue. This does not claim that knowledge must belong to a particular species. Any subject whose life depends on knowledge has reasons to care about understanding and influencing judgments that govern that life.

5. Postmodern nihilism: plural values and potentially uniform evaluation

5.1 The absence of a shared endpoint does not make everything worthless

Lyotard's discussion of the postmodern concerns distrust of grand narratives and the relationship between the legitimacy of knowledge and performance criteria. The present paper borrows this structure of questioning: when people do not share a predetermined historical destination, how can they justify different forms of engagement? It does not follow that everyone today rejects value, or that modern societies once enjoyed an uncontested common faith. Equating the postmodern with decadence reduces theory to a label for the mood of an era. [10]

Plural sources of value can expand freedom. Religion, intimate relationships, creativity, public service, or solitude may provide direction without universal agreement on a life template. The difficulty is that different reasons for living need not receive equal institutional support. Organizations may compare speed and output precisely while struggling to recognize careful judgment, slow learning, or concerns resistant to commodification. Individuals then face an asymmetry: they may choose their own meaning but still have to justify that choice through a narrow set of measurable criteria.

AI may intensify or relieve this asymmetry. It can lower barriers to knowledge and creativity, while extending comparisons of efficiency that make previously adequate effort appear insufficient. If saved time is immediately absorbed by new targets, people gain no space to choose a direction; they are required to follow a faster standard. This is a conditional claim about organizational rules, not evidence that the pattern is universal wherever the technology appears.

5.2 From lacking a purpose to being unable to trust one

A crisis of meaning need not take the form of not knowing what one wants. Someone may still want to conduct serious research, raise a child, or contribute to a community, yet increasingly doubt that commitment can bear a reliable relationship to outcomes. The goal persists while the route is repeatedly rewritten; past dedication remains while the social value assigned to it changes. Meaninglessness then has a temporal structure: it becomes difficult to connect accumulated experience, present action, and future expectations.

This interpretation explains why more options alone may be insufficient. Many choices, none of which appears durable, can encourage repeated experimentation without commitment. Commitment here does not mean refusing change. It means investing time, relationships, and energy that cannot immediately be recovered. When all commitments must remain endlessly adaptable, losses fall on those expected to adjust, while institutions can continue to claim that opportunities have never diminished.

5.3 Recognition must take the form of standing and relationships

Recognition is more than encouragement. Fraser directs attention to institutionalized standing and conditions of participation, while warning against substituting cultural affirmation for distributive questions. Applied to

AI-mediated work, assurances of irreplaceable human creativity do little if workers cannot influence goals, contest arrangements, or share gains. Conversely, an organization need not proclaim everyone exceptional to provide more substantive recognition: opinions must be able to affect decisions, and basic security must not depend on humiliating tests of worth. [11]

Recognition cannot mean equal praise for every contribution. Professional judgments vary in quality, conduct can be criticized, and allocation requires justification. The paper distinguishes equal respect that is not withdrawn for poor performance from differentiated assessments of particular contributions. Eliminating the latter risks removing public standards for doing things well. Making the former depend on performance risks depriving people of equal membership when their competitive advantage disappears.

5.4 Capabilities can spread without an equivalent spread of esteem

AI may allow more people to produce formerly scarce outcomes. This can expand opportunities without expanding esteem proportionately. Where evaluation depends on relative rank, improved performance by everyone can raise entry standards. A person's capabilities may genuinely increase while their self-assessment deteriorates: they do more, but the threshold for attention has risen too. This is a deduction about a competitive structure, not a general psychological effect measured in this paper.

Skills training alone is therefore an incomplete response. It may improve agency without changing arrangements in which a few ranks monopolize recognition. The deeper question is whether society can recognize actual contributions without positional advantage while retaining standards of quality. Eliminating evaluation is impractical; making dignity depend entirely on competition is unacceptable. An alternative must identify concrete value, allow plural contributions, and avoid withdrawing membership when someone fails.

5.5 When finding meaning becomes another compulsory assignment

A society can relax the demand for a single life trajectory while transferring the entire burden of explanation to individuals: unstable employment calls for a new identity, unreliable relationships for learning solitude, and an uncertain future for discovering a personal passion. Each suggestion can help in particular circumstances. Together, however, they can form another demanding standard. People must endure change and demonstrate that they have converted it into an affirmative, distinctive life story.

Meaning then shifts from a reason supporting life to an outcome one must deliver. Someone who loses direction may experience both the original loss and failure at the assignment of living meaningfully. This is proposed as a possible normative pressure, not a measured diagnosis of an era. Its structure is that society weakens the conditions for stable commitments while asking individuals to create sufficiently stable reasons alone. Difficulty doing so is attributed to deficient inner resources, recasting a distributive problem as a failure of self-management.

AI may enter this cycle by summarizing strengths, generating plans, simulating appreciation, or continually explaining why someone deserves to be needed. Such support can genuinely ease loneliness and should not be rejected merely because it is technological. Yet if the missing condition is standing to affect shared decisions, even precise positive feedback cannot substitute for that standing. Helping excluded people endure exclusion and reducing exclusion are different effects. They may coexist or become disconnected.

The difficulty is that individuals still need to make decisions about their own lives. Attending to institutions does not mean every choice can wait for structural change, or that avoidance should never be examined. Responsibility itself is not the target. The objection concerns concentrating responsibility on those with the least control. Individuals may attempt to rebuild their lives while organizations and public institutions must also explain why some people's commitments are so readily invalidated and who bears the costs afterward.

The openness of meaning therefore does not excuse society from explaining its rules. The absence of one purpose for life does not prevent scrutiny of evaluative arrangements that repeatedly disadvantage certain lives. An open society must permit different answers and also permit a further question: why must I use your standard to establish that the answer I have chosen deserves a place?

6. Four pathways from task displacement to a crisis of meaning

6.1 Income: the material conditions of the future become less dependable

The most direct pathway concerns income and protection. Lower demand for tasks can affect wages, job stability, and bargaining power, making housing, care, and long-term plans more difficult to arrange. Anxiety in this situation may arise entirely from material risks, without philosophical self-negation. If adequate and credible economic protection substantially relieves the difficulty, the material explanation should be acknowledged. A theory of meaning must not preserve itself by renaming every economic problem an existential crisis.

Equal income does not necessarily mean equal circumstances, however. A fixed allowance and a continuing remunerated social role may differ in predictability, contact with others, and self-understanding. Research must distinguish the stability and terms of support, including stigma, rather than assuming that replacement of an amount of money repairs every loss. Economic and recognition pathways can coexist, but their coexistence requires evidence rather than conceptual possibility alone.

6.2 Practice: more output, but less clarity about one's own contribution

The second pathway concerns practical agency. People need to understand what they have done, which outcomes they influence, and where they can change direction. Tools can extend this capacity or reduce someone to approving results they cannot understand. Agency does not require performing every step alone: doctors, engineers, and researchers have always depended on others' work. What matters is whether cooperation preserves understanding, discretion, and learning commensurate with a person's responsibilities.

Danaher and Nyholm's achievement-gap argument examines the link between valuable outcomes and human contributions. Scripter replies that losing particular opportunities for achievement need not imply a net loss of all such opportunities, and that autonomy requires separate consideration. The unresolved question matters here: the value of outcomes, individual contribution, and control of the process can diverge. More output does not establish greater human achievement, while less strenuous work does not establish its inevitable reduction. [12], [13]

Consider an explicit thought experiment involving two editors using the same AI system. One chooses topics, verifies materials, revises arguments, and rejects unsuitable suggestions. The other accepts batches of text under fixed targets and bears responsibility for errors without the authority to change the process. If their experiences of meaning differ, the label of AI use explains too little. Discretion, understanding, workload, or responsibility may account for the difference. This example reports no empirical result; it identifies variables that a meaningful comparison would need to distinguish.

6.3 Standing: from needing one's judgment to requiring one's compliance

The third pathway concerns social standing. Technology need not replace all of a worker's abilities to change how others consult them, seek their opinion, or place trust in them. Someone previously treated as an experienced member may become a supplementary operator retained until automation is complete. If this change persists, it can alter a person's understanding of their position even while income remains stable.

Being needed requires careful analysis. Dependence can provide meaning, but it can also enable exploitation. Organizations may invoke indispensability to demand unpaid overtime, while families may close off other possibilities by insisting that only one member can provide care. Meaning therefore cannot equal the degree to which others depend on me. Worthwhile relationships should permit negotiation, refusal, and substitution rather than require another person's helplessness as evidence of one's importance. Effective participation differs from being trapped in a role.

6.4 Narrative: the past no longer leads toward the future

The fourth pathway connects personal history with the future. Long training matters not only through current income but through an expectation it once supported: present effort would become the foundation of future

judgment, standing, and relationships. Rapid changes in technology and evaluative standards can unsettle that expectation. The loss concerns more than a readily priced sunk cost; it concerns an account of how a person became who they are.

Scripter's argument about the historical embeddedness of meaning helps explain why occupational losses cannot always be fully compensated through other activities. This paper extends that insight into a research question: can people still form credible long-term plans? If someone doubts that their own efforts can accumulate, they may also hesitate about a child's education, development, and future position. Demographic evidence must test that extension independently. Philosophical coherence cannot establish an effect on actual births. [14]

The four pathways can be organized by asking what technology changes in resources, agency, relationships, and expectations. They are not an inevitable sequence. Someone may experience income risk without a crisis of meaning; another may gain financially while losing ownership of achievements; another may benefit along several dimensions. Separating pathways allows the account to face these differences rather than create an apparently complete but untestable story of decline.

6.5 The paradox of being needed: do we want others to be unable to leave?

The wish to be needed can contain different aspirations. I may want my actions to help others, want others freely to choose my company, or want them to be unable to leave me. The first two can support relationships; the third can undermine another person's freedom. If meaning depends on others permanently lacking alternatives, my loss when technology reduces their dependence is entangled with their liberation.

Care makes the tension especially clear. Good care can help its recipient become more independent and require less assistance. A teacher may similarly regard a student's eventual independence as success. The value of contribution cannot therefore be measured simply by whether dependence continues. A practice can succeed by reducing the demand for itself without making its earlier significance disappear. Conversely, persistent demand might mean the underlying problem remains unresolved.

This does not entitle us to demand that contributors experience no loss. A carer may feel both satisfaction and emptiness when someone gains independence, and a worker may acknowledge a technology's benefits while mourning a disappearing role. These responses need not cancel each other. Requiring people to leave cheerfully as proof of selflessness mistakes attachment to particular relationships for a moral failure to support progress.

A more demanding standard is needed. People should have opportunities to contribute in ways that benefit others while permitting their freedom. When those contributions reduce the contributor's own necessity, basic support, social voice, and recognition of their history should not disappear with it. This separates a change in one's role from exclusion from membership. The former is often an ordinary feature of shared life; the latter requires an additional justification.

The central question consequently changes. People need not show that the world cannot do without them to have reasons to participate. Yet an institutional order that continually tells some people they are no longer needed cannot discharge its responsibilities merely by advising them to find hobbies. Meaning should rest neither on possessing other people's needs nor on treating acceptance of one's exclusion as a mark of maturity.

7. What the existing empirical evidence actually supports

7.1 The completed public-data reanalysis

The paper draws on an earlier secondary analysis of Lee and colleagues' public data. The original experiment reports 408 completers, of whom 269 instruction-compliant participants appear in the public main file: 119 wrote independently, 50 drafted first and then used AI for revision, and 100 copied AI output directly. Five original items were used to recalculate meaning scores on a 1–7 scale and check the supplied aggregates.

The other 139 completers are not included in this public file, preventing recovery of a full intention-to-treat estimate. Post-treatment selection also limits the protection offered by original random assignment. [15]

Mean meaningfulness in the initial task was 5.541, 5.464, and 4.944, respectively. Direct copying was 0.597 points lower than independent work, with a 95% marginal confidence interval of [−0.982, −0.212], Hedges' g=-0.423, and a Holm-adjusted p=.0076 across the three comparisons within that stage. In a separate survey of 270 respondents, passive use and active collaboration had Pearson correlations of .052 and .005 with general work meaningfulness; neither association was statistically significant. Methods, aggregate results, and access to the original material appear in the reproduction guide. The two samples are not pooled, and this is not a new independent experiment.

The warranted interpretation is specific: a prescribed mode of participation was associated with lower task meaningfulness, while the everyday-work survey did not show a similarly clear linear relationship. Copying simultaneously altered effort, judgment, ownership, and the experience of complying with instructions, so it does not isolate a single mechanism. Neither existential nihilism nor reproductive decisions were measured. Describing the result as AI causing humanity to lose meaning would exceed the evidence. Treating near-zero survey correlations as proof that no risk exists would also be unwarranted.

7.2 Assigning evidence the weight it can bear

The table separates the objects of inquiry so that studies at different levels are not assembled into an untested causal chain.

Material	What is actually observed or analyzed	What it helps assess	What it does not directly establish
Writing experiment and public-data reanalysis	Short-term task assessments under specified workflows	Participation modes and task experience should be distinguished	Occupational identity collapse, existential nihilism, or fertility effects
Everyday-work survey	Self-reported AI use and work assessments	Whether associations are apparent and where interpretation is limited	Long-term causality or an overall absence of effects
Demographic narrative experiment	Scenario information and immediate fertility intentions	Future narratives may influence intentions	Actual births or a mechanism specific to AI
Regional robot study	Technological adoption and regional fertility differences	Effects vary across regional and institutional contexts	Psychological mediation within each individual

Null findings change the question the paper should ask rather than remove its rationale. If a general association between greater AI use and lower meaning is unsupported, attention should turn toward practices, standing, and life narratives. That shift must not make the theory infinitely accommodating. If every result can retrospectively be attributed to context, the account cannot fail. A later section specifies observations that would require its claims to be narrowed.

8. From doubts about one's worth to hesitation about the next generation

8.1 Distinguishing demographic indicators

Discussion of declining birth rates must distinguish birth counts, crude birth rates, period total fertility rates, completed cohort fertility, fertility intentions, and ideal family size. Birth counts are affected by the size and age distribution of the reproductive-age population. Period total fertility rates can be affected by postponement. An ideal number of children is not the same psychological object as intending a birth within the

next few years. Delaying childbearing does not necessarily imply permanent childlessness, and a declining national birth count cannot be explained directly by a population-wide loss of meaning.

Temporal order also matters. The OECD reports a decline in its members' average total fertility rate from 3.3 in 1960 to 1.5 in 2022, discussing employment, housing, family arrangements, and changing norms. This long trajectory predates the widespread use of contemporary generative AI. Generative AI therefore cannot explain the entire history of low fertility. The proposed mechanism might intensify, alter, or relieve existing tendencies; it is not a recent single cause for decades of demographic change. [16]

8.2 Reproductive decisions involve judgments about a future world

Childbearing entails a long-term responsibility that cannot simply be withdrawn. People consider present affordability but also imagine whether they can provide reliable care, how responsibilities will be shared, and what society a child will encounter. The future is not an already existing fact awaiting calculation. People must form an understanding sufficient for action under incomplete knowledge. Vignoli and colleagues' Narrative Framework brings expectations, imagination, and socially communicated stories into this analysis. [17]

This does not make economic conditions unimportant. Income, housing, and childcare are direct constraints as well as clues to whether a future can be committed to. An insecure contract may affect mortgage eligibility, household divisions of labor, and judgments about long-term responsibility at once. An account of meaning adds nothing if it merely redescribes these factors in more evocative terms. It must identify narratives and relationships that matter when relevant resources are comparable.

Vignoli and colleagues conducted scenario experiments with 800 participants in Florence and 874 in Oslo. Positive and negative economic narratives were compared with a control condition receiving no narrative, followed by questions about intentions to have a child within three years. Narratives affected immediate intentions, with effects depending on context. This supports including interpretations of the future in fertility research. Participants came from particular cities and couple samples, however, and actual births were not followed. The study does not directly establish that AI-related crises of meaning lower population fertility. [18]

8.3 The specific hypothesis proposed here

The hypothesis is that when technological change weakens expectations of a stable place through work, while occupational success remains a criterion of responsible adulthood, people may become less confident both in their own long-term plans and in the circumstances awaiting the next generation. Under some conditions, this may lead them to postpone or forgo births they would otherwise want. The hypothesis contains separate links: work changes, people reassess their position, that reassessment enters their understanding of the future, and the future judgment affects reproductive plans.

Any link may fail. Some people understand AI as a prospect of greater resources and freedom for their children. Some turn toward family as a source of direction. Others never make occupational position central to self-evaluation. Partners' preferences can differ, and outcomes emerge from negotiation and constraints. The paper therefore does not assume a necessarily negative effect; direction, magnitude, and relevant populations remain empirical questions.

The most important issue is not a simplified story in which people feel useless and therefore avoid children. It is the continuity of a social promise: when education, effort, occupation, and security no longer appear reliably connected, do people believe they can bring a child into a world with real opportunities to participate? This concern may coexist with material costs and might remain after economic improvement, but the latter possibility requires additional evidence.

8.4 Supporting indications and counterexamples that must remain visible

The UNFPA–YouGov survey covering 14 countries provides indications of constraints. Its report states that 39% of respondents considered financial limitations to have affected, or potentially to affect, realization of

their desired family size. Nearly one in five reported concerns about war, environmental conditions, or other future risks as reasons for having, or expecting to have, fewer children than desired. These proportions must be understood within the survey's coverage, not as unconditional estimates for the world population. Self-reported reasons do not isolate AI or a crisis of meaning. The most relevant distinction is between voluntary choices and an inability to realize reproductive wishes. [19]

More direct technological evidence is not uniformly negative. Matysiak, Bellani, and Bogusz examine regional robot adoption and fertility in six European countries. They report relatively small effects that vary with industrial, educational, and technological conditions; some regions may benefit. Regional estimates do not establish whether individual households experience nihilism, and industrial robots are not synonymous with generative AI. The study challenges a simple inevitability: technological adoption need not have the same demographic consequences everywhere. [20]

A Chinese study using 2014, 2018, and 2022 CFPS data and regional AI patent indicators reports lower ideal numbers of children among women in relation to AI development, discussing employment, relative income, family interaction, and gender norms. The present paper checked the journal's summary and did not re-estimate the models. Notably, higher employment and income can increase the opportunity cost of childbearing and accompany lower intentions. That differs from the proposed mechanism of feeling no longer needed and cannot be counted as evidence for the same explanation. [21]

Doepke and colleagues' review of fertility economics shows why simple universal negative relationships among income, women's employment, and fertility are inadequate for high-income societies. The compatibility of careers and families is important. For this paper, fathers' involvement, care support, and work arrangements therefore warrant close attention. Women's increased occupational choice must not be described as a loss of meaning. Choosing fewer children when options expand can express greater agency. [22]

8.5 Childbearing is not an obligation to demonstrate meaning

Even if existential uncertainty affects some reproductive decisions, childbearing does not become a duty to cure meaninglessness. Children have interests and futures of their own. They cannot be treated as instruments for filling adult emptiness, maintaining labor supply, or demonstrating civilizational vitality. A person who chooses not to have children can have a profound, stable, future-oriented life; a parent can experience a crisis of meaning. Demographic indicators are not individual life-performance scores.

The paper accordingly rejects two symmetrical simplifications: lower fertility cannot all be interpreted as autonomous liberation, nor as social despair. Desires, constraints, and outcomes must be distinguished, allowing hesitation, care, and autonomous choice to coexist within one person. Institutional improvement should be assessed through the ability to choose childbearing or nonparenthood without coercion or unfair burdens, not solely through whether birth counts rise.

8.6 Why saving labor and wanting more children can coexist

Firms may seek to reduce the cost of particular jobs, public systems may need future contributors and carers, and households bear the immediate time and risks of raising children. These demands originate from different actors and operate across different periods and sectors. Saving labor and wishing for more births are therefore not a direct formal contradiction. Ageing can also create shortages in some sectors while other occupations face displacement. Treating all labor as instantly interchangeable creates an excessively tidy paradox.

The absence of a formal contradiction does not remove social conflict. If households are asked to bear costs for long-term population stability without credible support for housing, care, and the next generation's opportunities, collective goals may become disconnected from private feasibility. A society cannot invoke children's importance when it needs future workers while treating the time and circumstances of current carers as purely private concerns. This criticism concerns responsibility, not a duty for every household to meet population targets.

Moreover, if children matter primarily as future producers, taxpayers, or carers, the initial predicament is passed to the next generation: people are asked to prove future usefulness before they are born. Intergenerational meaning should include open-ended concern for people not yet present rather than an advance assignment of economic functions. Public consequences of population structures remain legitimate subjects, but discussion must leave room for different individual directions in life.

8.7 Why it matters that others will live after we die

An intergenerational future concerns more than the number of future workers. It also bears on how people understand their actions now. In "The Afterlife," Scheffler examines continuation in a nonreligious sense: others living after one's death. Thought experiments concerning catastrophe and the absence of future births explore how that expectation supports present commitments, while distinguishing confidence in value from value itself. This paper borrows the question; thought experiments are not fertility surveys. [23]

This starting point permits a further question of our own. Someone preserving local memories, maintaining a language, building a school, or protecting an ecosystem may care about beneficiaries beyond their own lifetime. Those beneficiaries need not include biological descendants. Concern for future generations cannot therefore be identified with a personal desire for children. People without children can participate deeply in a world that later people will inherit.

AI moves the question into new territory. If knowledge, stories, and skills can be stored by machines for long periods, is preservation equivalent to continuation? Consider a constructed world with complete archives but no one who understands their disputes, alongside a world with incomplete archives that later inhabitants continue to use and revise. Both preserve aspects of the past. One retains objects; the other also sustains a relationship of response. This comparison does not assume artificial subjects cannot understand. It asks what exactly we hope to preserve.

If future artificial subjects can understand, value, and inherit, the argument cannot simply retreat to biological boundaries. We must distinguish concern for the continuation of a species, a civilizational tradition, or subjects capable of experience, disagreement, and mutual responsibility. These concerns can overlap or conflict. Humanity's special attachment to its history can explain a concern's strength without automatically showing that it should override every other subject's interests.

Even great importance attached to later people does not directly assign reproductive duties to particular individuals. Hoping that a school, community, or language has a future does not authorize forcing one person to bear all the costs of continuation. Intergenerational concern should first call for improving a world worth inheriting, caring seriously for people already present, and sharing the risks of commitments to the future. Skipping those responsibilities and demanding reproduction in the name of civilization evades the contradiction examined here.

8.8 Must the next generation inherit a debt to demonstrate its worth?

An adult who repeatedly feels required to prove entitlement to a decent life may experience more than cost anxiety when considering children. A moral hesitation can arise: would I bring someone else into the same endless competition for standing? This is a possible reason deserving investigation, not one whose prevalence follows from its intelligibility. It also differs from judging all births undesirable. Someone may instead reject presenting an unjust arrangement of life as an unchangeable natural order.

The hesitation exposes two risks in reproductive narratives. One asks children to become sufficiently successful to vindicate their parents' sacrifices. Another asks children to answer their parents' meaninglessness. Both can impose adults' debts of justification on people whose own wishes have not yet developed. Children may transform the meaning of a parent's life, but that possible outcome cannot establish a right to make them follow a predetermined path.

Conversely, describing reproduction entirely as private consumption overlooks a real, lasting relationship. Raising a child is not purchasing a predictable experience. It involves living with someone gradually able

to disagree with oneself. Treating the next generation as open-ended subjects makes the commitment one of providing conditions before independence while accepting that they may reject our cherished answers. That commitment is less predictable than predetermined inheritance and cannot be exhausted by one calculation of costs and benefits.

Openness nevertheless does not remove practical responsibilities. An unpredictable future does not excuse ignoring existing conditions. Parents' inability to guarantee happiness does not excuse institutions from providing basic support for housing, care, and education. Between absolute guarantees and abandonment lie defensible standards: addressing foreseeable difficulties, refusing to describe avoidable harms as necessary character-building, and allowing later generations to help revise the rules.

The population question can therefore be stated more precisely. Does a society give people reasons to believe that the next generation will be able not merely to survive but to form lives without repeatedly having to prove entitlement to acceptance? An affirmative answer does not guarantee more births. A negative one does not establish the cause of every choice against parenthood. The question nevertheless moves demographic discussion beyond the required number of future workers toward the shared life we are willing to offer those who come after us.

9. Seven objections that cannot be avoided

9.1 If new jobs will emerge, is the problem exaggerated?

New work and demand may emerge, which is an important objection to technological fatalism. Aggregate jobs alone do not answer distributive questions, however. New opportunities in one industry and displaced workers in another may be separated by geography, skills, age, family responsibilities, and time. The account must admit aggregate improvement while examining who can enter new positions. If most affected people gain access to better practices at bearable cost, the scope of the crisis-of-meaning thesis should narrow accordingly.

9.2 Is this merely elite anxiety over lost advantages?

Some anxiety about human distinctiveness may indeed reflect established professional groups losing scarcity advantages. AI can weaken old barriers by enabling newcomers to express ideas, create, and solve problems. Protecting meaning must not automatically justify protecting incumbents. Loss of exclusive privilege should be distinguished from loss of basic participation: the former need not be unjust, while the latter warrants a response. The same technology may unsettle former elites and give many others real capabilities.

9.3 With income and leisure, why must anyone still be needed?

This objection challenges the centrality of work. If someone is content with a life independent of occupational identity, there is no reason to force them back into performance competition. Others do want to make a difference to people and shared affairs. Institutions can offer opportunities without requiring contribution to be everyone's central purpose. The boundary concerns opportunity and obligation: resisting systematic exclusion from consequential activity does not mean requiring continuous usefulness as a condition of support.

9.4 Could money and care costs explain all of it?

They may explain a substantial part. If secure income, affordable housing, and fair care arrangements account for reproductive changes, researchers should not enlarge their theory's territory by adding existential anxiety. A meaning pathway deserves retention only if it contributes after measurements, temporal order, and confounding are properly distinguished. Even then, it need not have the greatest policy importance. A concept's apparent depth does not establish the size of its contribution to observed change.

9.5 If AI performs even the most valuable activities better, is participation still real?

This is the strongest objection. O'Brien considers a future in which AI displaces intrinsically valuable research and creativity, questioning whether retreat into artificially structured games compensates for the

loss. Enjoying the process is not an automatic answer. Discovering a fact for the first time differs from repeating a known answer; helping someone differs from receiving a task designed merely to make one feel useful. If human activity no longer produces needed outcomes, some forms of achievement may indeed lose value. [24]

The objection nevertheless requires distinctions concerning where value is realized. The world having an answer does not mean a particular learner understands it. A superior performer does not remove every value of a group rehearsing together. A system completing acts of care does not automatically fulfill commitments within an existing relationship. Some value is realized in changes to particular participants and relationships, and cannot be entirely substituted by a final product. This establishes that losses need not be comprehensive; it does not prove that every possible AI future preserves sufficiently rich human meaning.

9.6 If happiness can be supplied, does reality still matter?

Push the objection further. Suppose a future system could continually provide satisfying experiences in which people felt their creativity appreciated, judgment respected, and relationships reciprocated. This is a stipulated thought-experiment capability, not a claim about current products. If the people involved were happier, why should it matter whether their judgments changed anything? The objection requires an explanation of what meaning demands beyond feeling good.

At least three cases differ. A tool can help someone understand and change actual circumstances. A person can knowingly enter a fictional activity and willingly invest in it, as in shared games or storytelling. Or a system can make someone believe they exercise real influence while prearranging every outcome. The latter two may both provide pleasure, but voluntary fiction and deception are different relationships. Calling both simulation conceals the distinction between informed participation and being misled.

Nor should the second case be assigned inferior meaning merely because it lacks an external productive purpose. Chess has humanly instituted rules, yet understanding, responding, and accepting defeat within those rules can be real practices. The value of human activity does not depend entirely on resolving scarcity outside the game. Relevant questions include whether participants can endorse the rules, whether their actions affect the shared activity, and whether they can leave or change their participation.

The third case exposes a risk. Where someone values helping others, understanding the world, or influencing decisions, continually supplying corresponding feelings without corresponding relationships may violate their own commitments. The problem is not insufficiently vivid experience. It is that the provider has decided on their behalf what counts as fulfillment. Happiness cannot eliminate every question about meaning, but appeals to reality should not discredit voluntary entertainment. What requires protection is the opportunity to understand and judge the relationship in which one participates.

9.7 Preserving activities for humans: a justified entitlement or an expensive preference?

If machines can perform a necessary activity at lower cost, who should bear the expense of retaining human work for achievement's sake? Where the price is inferior care for patients or waste of public resources, the desire for achievement cannot simply take priority. Interpreting every instance of automation as deprivation would remove the interests of those served from consideration.

Yet a more efficient result must itself be specified. Some activities aim to develop participants' capabilities, express their views, and establish trust, rather than only deliver an external product. Comparing education solely by the correctness of a final answer can omit the purpose of learning. Comparing public deliberation solely by the speed of producing proposals can omit the standing of affected people in governing their affairs. Including those aims does not covertly weaken efficiency standards. It first identifies the activity being undertaken.

This cannot justify every reservation of activities for humans. We must distinguish conditions for basic life and participation, reasonable support for valuable learning and practice, and restrictions on tools designed

to preserve an incumbent group's advantage. The first two may merit collective support; the third calls for stronger scrutiny. Occupational labels cannot settle the classification. Beneficiaries, cost bearers, and alternatives must still be considered.

A defensible principle is that people should not lose basic standing when they lose a competition, but cannot require every domain to preserve rules that ensure their victory. Society likewise cannot treat faster output as a sufficient reason to remove every opportunity for participation. The principle leaves genuine conflicts: some practices should change, some opportunities warrant costs to preserve, and some losses remain uncompensated. Acknowledging these conflicts comes closer to the problem than promising that technology and human meaning can always coexist without sacrifice.

10. Where human meaning lies: a limited answer with real commitments

10.1 Standing as a person cannot be a prize in a performance competition

We must first ask why someone has to prove that they deserve to exist. Demonstrating competence for a particular job is reasonable, and decisions affecting others require justification. Extending these demands to a person's entitlement to exist, however, makes respect conditional for people with fewer capabilities, those temporarily unemployed, and those needing care. The threshold would also rise whenever stronger competitors appeared, without anyone being able to guarantee lifelong compliance. Making basic standing depend on relative ability leaves everyone's security contingent on temporary advantage.

The paper adopts a normative commitment to equal respect: people's interests, suffering, and voices deserve serious consideration, and they should not be excluded merely because production no longer requires their labor. This is justified not only by everyone's potential vulnerability. If public rules claim legitimacy for everyone affected, they cannot refuse justification to someone solely because that person is insufficiently useful to others. This is not a statistical finding but a political and ethical commitment requiring public defense. Different traditions may provide different grounds; the paper does not claim to resolve every foundational moral dispute.

The commitment guarantees neither equal praise nor the permanent preservation of every occupation. It protects a standing in distribution and collective decision-making that prevents people from being treated as mere leftovers. Someone without exceptional achievements, or temporarily unable to articulate goals, still has reasons to receive care, be heard, and participate in decisions affecting them. Separating this standing from assessments of outcomes is the starting point for preventing a discussion of meaning from becoming another performance test.

10.2 Dignity can persist while direction is absent

Equal respect does not solve every existential difficulty. Someone can be adequately supported yet have no reason they can identify for getting up tomorrow. Society should not assign a single answer, but neither should it return the entire difficulty to the individual. Experienced meaning requires concrete connections: what one cares about, what one is willing to undertake, and how action enters a world worth understanding and sustaining. Wolf's account of subjective engagement meeting worthy objects provides resources for this question. It is a starting point for discussion, not an uncontested final formula. [25]

Why is liking something insufficient to establish meaning? People can be devoted to harming, dominating, or deceiving others. Why is an external declaration of greatness insufficient? Someone forced into a noble cause may be unable to own that life as theirs. Meaning involves a relationship between personal engagement and defensible value, without requiring every activity to produce grand public outcomes. Friendship, understanding a subject, ordinary care, and modest creation can have serious reasons behind them.

The evaluative standard remains contestable. Who determines what is worth caring about? Might marginalized practices again be rejected by dominant groups? Those involved must therefore be able to offer reasons, face objections from people affected, and allow several forms of life to coexist. Society cannot monopolize a list of meaningful activities and require people to re-enroll in it as a new form of employment.

10.3 A position in a relationship matters without exclusive performance

People need not be the world's best at a skill to matter within a relationship or practice. Meaning often occurs in particular positions: this friend heard my difficulty, this collaboration changed our understanding, this action responded to an actual need. Such a position does not imply absolute logical irreplaceability. Relationships can end and people can leave. Once a relationship exists, however, commitments, shared history, and mutual understanding do not automatically become worthless when a more efficient substitute appears.

Nor does the argument assume that AI can never form relationships. If future artificial subjects can care, commit, and have interests, ethical discussion must expand accordingly. Reasons for human meaning should not depend on denying other subjects possible value. Respect for a person can coexist with respect for another kind of subject, just as differences in ability among people do not require excluding less capable people from the domain of value.

The danger is that relationships themselves might be evaluated entirely through efficiency. If companionship means only making distress stop quickly, education means only delivering correct answers, and public deliberation means only producing an optimal plan, the standing of those involved as participants may shrink. Better outcomes matter, but opportunities to understand, consent, object, and change collective activity are themselves among the things outcomes should reflect. This is a question of public justification that superior prediction alone cannot settle.

10.4 Meaning must permit rest, dependence, and failure

Responding to meaninglessness through participation risks creating a new demand: remain active, creative, and contributing, or be judged not to be living seriously. Such a demand again excludes people who are exhausted, ill, disabled, or losing direction. Participation here therefore includes receiving care, being present with others, recovering slowly, and temporarily withdrawing. It is not continuous delivery of visible outputs, nor a requirement to turn all private life into public contribution.

Failure does not automatically eliminate meaning. An inquiry that discovers no new knowledge may still develop understanding and relationships; care that cannot prevent death may still respond to the person's needs. Recognizing value in failure does not make every failure worthwhile. A resource-consuming project without a defensible aim does not acquire meaning merely through heavy investment. People need to reconsider, revise, and sometimes end commitments rather than continue solely to prove that earlier efforts were not wasted.

Meaning is thus neither a certificate secured permanently nor a ranking refreshed every day. It reflects human finitude: we cannot accomplish everything worthwhile or eliminate every risk. We form particular commitments in particular circumstances and answer for their consequences. Finitude does not make them false, and local value is still value. Yet some losses can exceed a person's capacity for repair at a given time. Society must allow that unfinished condition to exist.

10.5 What if there is still no ultimate purpose?

Asking where human meaning lies can also mean asking why the universe needs humanity. This paper has no evidence establishing that the universe has assigned humans a purpose, and AI development alone cannot establish that the universe is devoid of value. Binding this ultimate question to everyday meaning makes every concrete life wait for an external authorization that may never be available.

The limited answer offered here is that people need not prove themselves indispensable to an external system before having reasons to relieve suffering, seek understanding, establish trust, and participate in a shared world. Reasons for value can be offered and examined within those relationships without amounting to a universal human mission. Some will still find this insufficiently expansive. Religious and other philosophical traditions remain relevant for those seeking transcendent grounds. An open paper must explain the reach of its argument rather than use an attractive sentence to pretend the question has ended.

10.6 If an outcome is replaceable, is the life that occurred replaceable too?

The claim that a machine can do something too sometimes identifies genuine substitution and sometimes confuses the objects of an activity's value. It helps to distinguish four aspects provisionally: the result produced, what participants develop through the process, how they respond to one another, and how the activity enters subsequent life. These are not four unrelated kinds of meaning. They are aspects of one activity that may be affected differently.

Imagine a family preparing a meal together. A better system might provide a more balanced meal faster, removing burdens of shopping and cooking. Yet cooking together could also involve a child first learning to care for others, negotiating tastes, and sharing the day's experiences. Replacing the finished meal does not necessarily replace those processes. This is no argument for preserving all domestic labor. Where one person is continually compelled to perform unpaid work, appeals to shared memories can conceal unfairness. The actual relationships must be specified before judging what is saved and lost.

Human meaning therefore need not depend entirely on producing results the world has never contained. Understanding a proof already known to others still changes this learner's world. Listening carefully to a difficulty already heard elsewhere may still change this relationship. The absence of global originality does not make such changes fictitious. Reducing every valuable activity to a race to produce scarce discoveries misdescribes much ordinary understanding, cooperation, and care.

An objection remains: if artificial systems can also participate in these processes, perhaps helping people develop more effectively, why prefer humans? No universal priority rule is offered here. In some settings, assistance will enlarge relationships; elsewhere, a particular shared history will give people reasons to continue together. The central limits run both ways: replaceable outcomes do not establish that the associated lives are entirely superfluous, while valuable processes do not justify preserving every process regardless of cost.

10.7 If people form meaning, is it merely consolation?

A more radical objection challenges any limited answer. If the universe assigns no purpose, are understanding, care, and creation worth pursuing only because people say so? Might meaning be an interpretation invented to make life bearable? The objection requires a distinction. A reason's dependence on a human world does not imply that it can change arbitrarily without constraints.

A promise provides an example. It exists only in relationships capable of understanding and responding to promises, rather than as a natural object preceding every person. Once made, however, its breach and consequences cannot be judged solely by the promisor's present feelings. Value can arise through relationships while making demands on those involved. It need not first be shown to exist independently of all life before it can support criticism and responsibility.

Likewise, concern for suffering has biological and social origins without making inflicting and relieving suffering equivalent in every relevant respect. Explaining how a concern forms does not complete an assessment of whether it merits support. AI might help explain human emotional mechanisms and thereby change self-understanding, but explanation alone does not determine how those emotions should be treated. Describing something as merely a mechanism cannot erase the distance between explanations of origins and normative judgment.

Relationships can also generate cruel norms. Collective formation cannot itself guarantee justification. Whether an account of meaning depends on oppression, permits objection, or conceals avoidable harm requires continuing scrutiny. Equal respect is not presented as the only answer deducible from the universe. It is a commitment exposed to public challenge. Its force lies in explaining why affected people should be treated as subjects capable of offering reasons, rather than claiming exemption from the need for reasons.

This position may not satisfy every desire for final certainty. But lacking absolute certainty does not make all reasons for living equally weak. We can compare whether a relationship involves deception, whether an action responds to an actual need, or whether an institution permits correction, and revise commitments

accordingly. Meaning's dependability can arise partly through continuing responsiveness to reality rather than resting entirely on an immutable, uncheckable ultimate authorization.

10.8 Why freedom cannot mean continually starting over

If AI lowers barriers to learning, creation, and changing activities, people may gain more opportunities to begin again. That is valuable. A life that can be sustained only by continually beginning again, however, may struggle to acquire continuity. Commitments provide direction partly by relating present action to previous understanding and future expectations. If all accumulated experience is expected to become invalid at any moment, freedom may increasingly feel like an inability to remain anywhere.

This does not require lifelong fidelity to an initial choice. Relationships without exit and occupational identities that cannot be revised can also oppress. The distinction is between an ability to change and a continuous requirement to change. One expands choice; the other repeatedly transfers adjustment costs to individuals. Choosing a new occupation and being required to prove oneself anew whenever a stable practice emerges should not be covered by the same language of flexibility.

Stability need not mean rejecting novelty. It can mean recognition of past contributions despite changing tasks, time to learn and effective participation despite changing knowledge, or acknowledgment that a commitment mattered even after a relationship ends. Such continuity permits change without requiring someone to announce that their previous self was worthless in order to adapt to tomorrow.

Meaning may therefore require neither a permanently fixed identity nor indefinitely revocable choices, but continuity that can be revised. People should be able to say that they no longer live in the same way while still understanding how they arrived here. Individuals cannot supply this understanding entirely through storytelling. Recognition of earlier contributions, resources for transition, and continuing relationships affect whether the story has a credible basis in reality.

10.9 Finitude and the wish to leave a trace

The wish that work, creations, or children continue after one's death responds partly to life's finitude. Yet making meaning depend entirely on being remembered forever would condemn most particular lives to failure in advance. No generation can guarantee indefinite preservation of its names, works, or standards of value. Greater archival capacity cannot alone guarantee that later subjects continue to care about every person recorded.

This does not make the desire to leave traces childish. It requires distinguishing duration from value. An act of assistance may have a brief effect while genuinely changing someone's circumstances at that time. Something preserved for centuries may become irrelevant to later lives. An action need not become a permanent monument to deserve attention. Understanding and care cannot be judged never to have mattered merely because they will eventually end.

Nor should death and suffering be romanticized as indispensable sources of meaning. If longer lives and less disease become possible, profundity alone supplies no reason to reject them. Finitude reminds us that commitments involve choices; it does not show that greater deprivation or suffering makes life more meaningful. What merits preservation is the capacity to live seriously within limited conditions, rather than keeping those conditions permanently difficult.

11. What institutions can do, and what they cannot promise

11.1 Support should not depend on finding meaning first

If meaning requires time and security, institutions should not demand prior demonstrations of optimism or adaptability before providing basic support. Income protection, affordable public services, and transition resources deserve consideration because they expand real choices. Choosing among basic income, insurance, services, or other arrangements requires comparison of financing, coverage, fairness, and implementation. No particular policy is presented here as a verified treatment for nihilism.

Economic protection is also not everything. A person receiving support may still experience diminished standing if treated as a transition cost without a voice. Resources and decision-making therefore belong in the same discussion: who influences how AI enters work, bears mistakes, receives gains, and contests unreasonable arrangements? Reducing these issues to psychological adjustment makes affected people responsible for accepting change without allowing them to help determine its form.

11.2 Real participation requires more than an approval button

Organizations may point to a human remaining in the process as evidence of continuing agency. Presence is not necessarily substantive. When someone lacks time to verify results, cannot understand key grounds, or cannot reject outputs, their final signature may merely transfer responsibility. Effective participation should be assessed by whether disagreement can change the process, whether learning and understanding are supported, and whether responsibility corresponds to actual control.

Efficiency and safety can conflict with these aims. Rejecting a tool that reduces serious errors merely to preserve a person's sense of achievement can harm those served. Transferring all authority for efficiency can undermine learning and accountability. Institutions must compare these costs within particular domains rather than require either universal manual performance or automation of everything technically possible. Human participation merits protection, but the reasons and methods must withstand scrutiny by people affected.

11.3 Recognizing care must not reassign it to those with less power

Encouraging people to find meaning in family and community after labor demand changes can sound benign while treating care as the natural destination of women, low-income people, or displaced workers. Care is valuable, but its value cannot excuse the absence of pay, rest, occupational choice, and public support. If some people must compensate for the social costs of technological transition through unpaid care, reconstructing meaning becomes a justification for an unequal division of labor.

Reproductive policy should likewise avoid treating people as agents of demographic targets. Childcare, housing, working time, and fathers' responsibilities matter because they enable people to pursue their own plans, not because every recipient must have children. Persistently low birth rates after improvement do not alone establish policy failure; fuller voluntary choice may itself improve lives. At the same time, ageing societies must organize care and resources without using moral appeals to transfer responsibility to a small number of families.

11.4 Shorter working time and a shared world

Productivity can make reductions in necessary labor possible without automatically creating discretionary time. If organizations convert every saving into higher targets, people still lack room for relationships, learning, and public life. Working time, distribution of gains, and public spaces therefore belong to the meaning debate. Community, educational, and cultural activities also require dependable support rather than willingness alone.

Institutions cannot guarantee that everyone experiences meaning, and should not manufacture fictitious tasks merely to keep people busy. They can reduce systematic exclusion, provide accessible practices, and make room for discussing different values. Specific commitments still involve personal and relational uncertainty. This limit does not remove institutional responsibilities; it requires abandoning the ambition to produce uniform happiness or uniform meaning.

11.5 Three possible AI societies: provision, competition, and shared decision-making

Three explicitly fictional future settings help concentrate the institutional question. They assume similar technological capabilities and distributable resources while differing in recognition and decision-making. They do not predict which future will occur. They show how comparable productivity can coexist with different human circumstances.

The first society provides basic consumption to displaced people but assigns important decisions to a small group of administrators. Residents can live safely and enjoy leisure while having little influence over conditions of support or reliable opportunities to challenge how systems organize their lives. Provision itself is not the problem; provision and dependence coexist. A person may no longer need to demonstrate deservingness for food while still having to demonstrate sufficient compliance to retain a protected position.

The second society primarily converts technological gains into new competitions. Tools are widely available, while learning speed, creative volume, and influence become continually rising standards. Individuals appear to have boundless choices yet struggle to determine when they have done enough because everyone's performance changes the benchmark. Equalized capability does not automatically produce security in evaluation. If the rules continually select a small group of winners, growing output cannot eliminate the position of those left behind.

The third society attempts to connect material protection, accessible practices, and shared decisions. It promises no universal excellence, but enables affected people to discuss how technology is introduced, saved time distributed, and common activities sustained. This is no conflict-free utopia: deliberation may be slow, incumbents may exclude newcomers, and participation may be merely formal. Actual influence, access, and correction must be examined. Procedural labels alone cannot establish meaningful participation.

These settings clarify that the institutional aim is not for a state or platform to assign everyone a role that feels important. It is to prevent any actor from monopolizing the definition of meaningful life. Society can refrain from offering one answer while remaining accountable for how participation is distributed. Public support should concern the conditions for forming and revising reasons, rather than merely producing satisfaction with existing arrangements.

11.6 Mourning in transition: loss need not immediately become growth

When an occupation, skill, or community declines, people can lose more than income: familiar ways of judging, trusted identities, and longstanding relationships may also disappear. Acknowledging aggregate technological benefits does not require immediately rewriting these losses as stories of growth. Treating every expression of grief as resistance to progress deprives society of the ability to hear the costs of transition.

Mourning here does not demand permanent occupational protection. It acknowledges that invested lives occurred and should not retrospectively be declared useless because demand changes. Commemoration, continuing peer relationships, transmission of experience, and voluntary transitional arrangements might allow the past to enter the future rather than simply be removed by it. Their effects require observation, but their justification need not depend entirely on immediate improvements in re-employment efficiency.

Recognition of loss should not fix people permanently in the identity of a victim either. Support must permit continuation, change, and exit while respecting different interpretations of personal history. Some people want to preserve a professional community; others finally wish to leave work they never enjoyed. Writing everyone into the same account of loss would repeat the very imposition of life definitions that the intervention seeks to correct.

12. Turning the argument into research that could fail

12.1 Three revisable propositions

First, under comparable AI task exposure, changes in discretion and perceived ownership may explain changes in work meaning better than general frequency of use. If reliable repeated measurements reveal no additional explanatory value, or if income and workload fully account for the relationship, the practical-agency pathway should be narrowed.

Second, changes in work meaning may enter overall life assessments more readily when occupational identity is central and other supportive relationships are weaker. If effects are unrelated to those conditions, or if people generally recover equivalent meaning rapidly after occupational disruption, the interpretation

emphasizing identity and history requires revision. Conversely, individual accounts of suffering do not establish a general effect.

Third, expectations about one's own and the next generation's opportunities to participate may influence reproductive intentions after wishes, material circumstances, and care arrangements have been properly distinguished. If the association is unstable, or merely reflects pre-existing intentions shaping accounts of the future, an independent meaning mechanism should not be asserted. Even a demonstrated intention effect requires separate follow-up of births and timing.

12.2 What should be observed

One feasible approach follows the same people before and after AI introduction, recording tasks, income, control, meaning, and future plans while distinguishing actual individual use, organizational arrangements, and occupational exposure. People who change jobs, become unemployed, or leave the labor market must remain observable rather than restricting analysis to successful stayers. Current-work meaningfulness cannot simply be set to zero after unemployment, nor should those respondents be dropped. Life meaning, material circumstances, and relationships should continue to be measured.

Partner-level information is needed because one person's wishes often do not determine reproduction. Ideal family size, near-term intentions, attempts, and births should be distinguished, alongside existing children, age, health, housing, and care arrangements. These variables cannot simply all be inserted into a regression. Income and future expectations may be mediators as well as consequences of shared causes. Inappropriate adjustment can remove the very effect of interest or introduce selection bias. Causal questions and temporal order must therefore precede the choice of comparison.

Thought experiments could inform studies varying organizational participation conditions, but their outcomes would still be immediate responses to scenarios. Researchers should not create fear through false personalized unemployment predictions, nor make increased fertility intentions an intervention goal for participants. Recruitment, ethical review, measurement adaptation, and sample-size planning would all need to precede implementation. None of that work has been undertaken in this paper.

12.3 What counterevidence is required

Fertility changes in regions with high AI adoption do not establish an AI effect because housing, education, migration, and industrial structure may change together. Population ageing and labor shortages may also encourage automation in the reverse direction. A one-time questionnaire cannot readily establish whether pessimism reduces reproductive intentions or existing intentions shape descriptions of the future. Serious inquiry must actively investigate these directions rather than collect only correlations consistent with the account.

Qualitative research should also include counterexamples: people liberated by displacement, people who lose work meaning yet still want children, people without occupational crises who choose not to reproduce, and people whose new AI-enabled capabilities strengthen their confidence in the future. Interviews should start from specific events, decisions, and relationship changes rather than ask whether AI makes life meaningless. Otherwise, researchers risk teaching participants a grand narrative and then treating their responses as a discovery.

12.4 Turning deeper questions into narrower comparisons

The added philosophical distinctions should affect the questions asked in subsequent research. Feeling appreciated should be distinguished from actual influence over decisions, freely chosen relationships, and praise without substantive effects. Organizations with comparable support but different arrangements of participation could be compared. If actual influence adds nothing to accounts of life meaning, explanations based on standing and participation would require further qualification.

Intergenerational inquiry should also distinguish personal reproductive plans from public concern for later people. Someone without a wish for children may make lasting commitments to education, environments,

and communities; someone who wants children may distrust the public future. Without these distinctions, research risks treating the number of children as a measure of concern for the future. The proposed deepening therefore makes comparisons more specific rather than asking meaning to absorb an expanding set of indistinguishable explanations.

13. An open conclusion: will society still make room for people?

The paper began with a contradiction: society may need less of certain kinds of labor while still asking people to establish their worth through work. The distinctions developed here locate the problem more precisely. It concerns not merely being outperformed but the possible weakening of income, participation, recognition, and commitments to the future. AI can extend human capabilities or close particular practices. Direction depends on technology, institutions, and relationships in combination, and should not be predetermined as a story of civilizational decline.

For population questions, the retained proposition is conditional. When people doubt that effort leads toward a stable place and that the next generation will have lives worth committing to, those judgments may affect births they would otherwise want. Existing evidence permits the question to be formulated and refined; it does not explain all low fertility or classify childless lives as nihilistic. Protecting human meaning and reproductive autonomy must remain compatible within the same argument.

How can people establish their meaning? The answer has two levels. As subjects deserving respect, people should not continually apply for permission to exist through productive performance. As people living particular lives, they still need to form reasons through understanding, relationships, creativity, care, or other worthwhile activities. The first level demands equal standing; the second permits difference, failure, and unfinished lives. Conflating them makes dignity conditional while making meaning empty.

Further scrutiny also prevents equating meaning with being needed, praised, or permanently remembered. A valuable contribution may reduce others' dependence, real participation can include failure, and cherished relationships will change. What requires protection is the ability to understand one's history, influence shared arrangements, and form new commitments amid change, rather than a guarantee of permanent irreplaceability. Intergenerational concern should begin here too: later people deserve space to form their own lives rather than a duty to prove that the preceding generation did not live in vain.

The hardest questions remain open. If future artificial subjects perform almost every valuable discovery and action, will sufficiently rich human participation remain? If an affirmative answer depends on institutions deliberately preserving a place for people, when is that place a justified entitlement and when is it an expensive symbolic arrangement? If some losses cannot be repaired, how should they be shared fairly without requiring those affected to prove their adaptation through optimism? A single argument cannot settle these questions.

What the paper ultimately asks is how shared life can continue to include particular people as productive capacity grows. Will society turn saved labor into time people can choose how to use, gains into shareable resources, and affected people's judgments into actual influences on decisions? Will someone who temporarily cannot explain why they are living still retain a place in which to live? Meaning need not come from a predetermined social answer. Yet a society that allows people to seek, lose, and reform their answers is itself a worthwhile common undertaking.

Limitations and an open mode of discussion

This is a working paper in social theory and normative argument. Literature was selected around a problem rather than through systematic sampling of all research. The value of its cross-disciplinary connections still requires fuller scholarly discussion. The existing empirical reanalysis covers short-term tasks and a cross-sectional survey, not the complete proposed mechanism. Demographic studies differ in period, country, technology, and measurement, and cannot be combined into a single identified causal effect. The proposed pathways may overlap, and their relative weights have not been estimated. Scenarios added in version 0.4.0

concerning future societies, capability dependence, and simulated experience are argumentative devices, not forecasts, interview findings, or new experimental results.

The postmodern is used interpretively rather than operationalized as a population-level psychological variable. Equal respect and substantive participation are normative positions rather than statistical findings. In particular, the paper establishes neither inevitable technological replacement, nor a causal explanation of low fertility through meaning crises, nor a policy that restores every loss. These limits do not remove argumentative responsibilities: the concepts, inferences, omitted literature, and counterexamples remain open to criticism.

The public version uses a common article identifier and bilingual version number, accept corrections, additional evidence, and objections directed at specific passages, and preserve a revision record. Contributors may challenge a pathway, offer different grounds of value, or present contrary cases without first accepting the paper's position. Open discussion is not external peer review, and GitHub publication is not journal acceptance. This edition provides the bilingual final artifacts within the author-confirmed scope; the paper does not recruit participants.

Materials, licensing, and AI disclosure

The earlier data reanalysis uses Lee and colleagues' OSF project 9qxjz. This version draws on completed scoring, Welch comparisons, Holm adjustments, and correlation analyses without adding individual data. The five-item meaning scores, sample selection, and principal results are described in Section 7. Complete computational methods, code, evidence records, and aggregates are maintained in the local working project; verification requests may be submitted through this paper's repository Issues. Original participant records and third-party article full texts are excluded from the public artifact package. The OSF page does not provide a clearly separate data license; public readability does not authorize relicensing someone else's data.

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